A T8T-SRAM Computing-in-Memory Macro for Ternary Deep Neural Networks and Boolean Logic Computations

Chenghu Dai, *Member, IEEE*, Jianhao Zhang, Zihua Ren, Lian Liu, Lei Meng, Chaoyi Wang, Chunyu Peng, *Senior Member, IEEE*, Wenjuan Lu, *Member, IEEE*, Zhiting Lin, *Senior Member, IEEE*, and Xiulong Wu, *Senior Member, IEEE*

***Abstract*—Deep neural networks (DNNs) play important roles in artificial intelligence applications and show hungry computility and power demands. Compared with binary neural networks (BNNs), ternary neural networks (TNNs) have higher representation and adaptive abilities and balance the inference accuracy and computing efficiency between DNNs and BNNs. This article proposed a T8T-SRAM computing-in-memory (CIM) macro to achieve Boolean logic operations and MAC operation of ternary activation and ternary weight. The proposed T8T-SRAM bitcell has a separate read and write path, and can avoid the read disturb issue. In Boolean logic operation mode, the T8T-SRAM macro can achieve NAND, NOR, XNOR, and XOR operations with redundant rows, reducing the additional reference voltage generation circuit. In the MAC mode, the result is quantized by an embedded column analog-to-digital converter (ADC), which uses activation refresh to reduce weight changing. In 28-nm CMOS technology, under 0.5-V array supply voltage and 0.9-V peripheral supply voltage, simulation results manifest that the MAC results have good linearity, and feasibility of Boolean logic operation. The proposed T8T-SRAM macro realizes MAC operation of 16 ternary activations and 16 ternary weights with 333.99–816.1-TOPS/W energy efficiency and 61.9-TOPS/mm^2 area efficiency. Using an ResNet-18 network for the inference of MNIST, and CIFAR-10 datasets, the accuracies were 99.06% and 85.76% with a ternary activation and ternary weight.**

***Index Terms*—Boolean logic, computing-in-memory (CIM), multiply and accumulate (MAC), SRAM, ternary neural network (TNN).**

I. Introduction

DEEP neural networks (DNNs) play important roles in artificial intelligence applications, such as natural language processing, automatic driving, and image recognition [1], [2]. For DNN processing, large amounts of data need collection and manipulated timely. In the traditional von Neumann computing architecture, the separation of the computing unit and storage unit leads to bandwidth limitations and low efficiency due to the frequent data exchange. The computing-in-memory (CIM) architecture is a potential method to overcome the von Neumann bottleneck, which combines computing unit and storage unit within memory banks [3], [4]. The CIM reduces the data exchange between memory and processors, thus considerably improving data throughput and energy efficiency [5], [6], [7].

However, the arithmetic complexity and memory access have limited the energy efficiency and acceleration of DNN hardware. Using low-precision activations and weights can improve DNN inference efficiency and reduce energy consumption. In recent works, weights and activations are binarized to +1 or −1, i.e., the binary neural networks (BNNs). To reduce the inference accuracy loss of binary multiplication, relevant studies proposed ternary neural networks (TNNs). Compared with binary DNNs, TNNs have higher representation and adaptive abilities, because their neurons can represent more states. TNNs achieve a good balance between inference accuracy and computing efficiency [8], [9].

It is foundational of the Boolean logic in computer science for its use in conditional statements, search queries, and electronic circuits. In addition, BNN and TBN significantly reduce the operation complexity by replacing multiply-and-accumulate (MAC) operations with Boolean logics such as XNOR, bitcount, or AND [10]. Integrating Boolean logic operations with MAC operations in a single-CIM macro can enhance the feasibility of the CIM macro.

In prior SRAM-based CIM studies, researchers have taken advantage of the storage characteristics of SRAM to process BNNs and TNNs. In a typical SRAM CIM macro, weights are stored in SRAM, activations are performed on wordlines by a digital-to-analog converter (DAC), and the conversion of CIM results to digital values is achieved by an analog-to-digital converter (ADC) [see Fig. 1(a)].

Ando et al. [11] proposed a reconfigurable accelerator architecture and significantly improve energy efficiency in process BNNs and TNNs. Yin et al. [12] presented an XNOR-SRAM macro for processing BNNS and TNNS. The XNOR-SRAM macro can activate all 256 computational rows simultaneously, and the MAC result is digitized by an optimized 11-level flash ADC embedded in the periphery. Jain et al. [13] proposed

Received 2 October 2025; revised 1 December 2025 and 25 December 2025; accepted 28 December 2025. Date of publication 12 January 2026; date of current version 27 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62274001, Grant 92464203, and Grant 62471003; and in part by the University Synergy Innovation Program of Anhui Province under Grant GXXT-2023-013. *(Corresponding author: Wenjuan Lu.)*

The authors are with the School of Integrated Circuits and Anhui Provincial High-Performance Integrated Circuit Engineering Research Center, Anhui University, Hefei 230601, China (e-mail: wjlu@ahu.edu.cn).

Digital Object Identifier 10.1109/TVLSI.2025.3650162

1063-8210 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:37:55 UTC from IEEE Xplore. Restrictions apply.

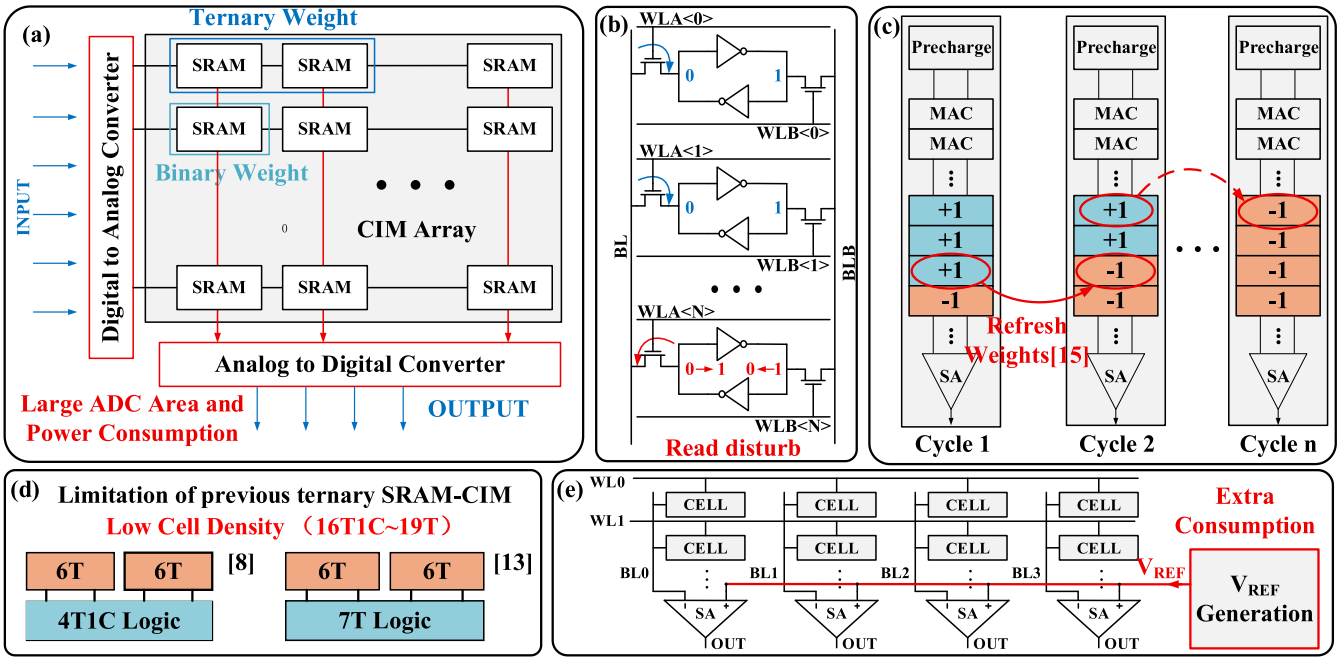


Fig. 1. Challenges in the previous ternary value CIM architecture. (a) Overhead of bitcell and ADC. (b) Read disturb. (c) Issue with the previously embedded column ADC. (d) Area overhead of the ternary weight storage. (e) Area overhead to process the Boolean logic operation.

a programmable in-memory accelerator TiM-DNN, which can implement both symmetric weighting operations ($-a$, 0, a) and asymmetric weighting operations ($-a$, 0, b) through 19T-SRAM. Jeong et al. [8] presented a 16T1C bitcell and charge-based psum adders (CPAs) to process TNN with good linearity. Wang et al. [9] used a dual-six transistor (dual-6T) SRAM cell with split wordlines to realize ternary weight and ternary activation. The proposed dual-6T memory array achieves up to 635-TOPS/W energy efficiency at 100 MHz.

However, the "read disturb" problem in dual-6T SRAM still exists as conventional 6T-SRAM. In addition, to alleviate the "read disturb" problem, 10T-SRAM and dual 7T-SRAMs [10], [14] are used for cases of ternary weight and ternary activation or binary activation and ternary weight.

Though previous works have promoted developments of the CIM macro for ternary DNNs, there are still some limitations (see Fig. 1).

1) For the ternary value bitcell, the ternary value SRAM bitcell still lacks a standard bitcell. Due to the multirows are opened at the same time, when the CIM macro uses dual-6T SRAM bitcell, which has the same read and write path, there are "read disturb" problems. And 19T-SRAM and 16T1C bitcell show large area overhead [see Fig. 1(b) and (d)].
2) For ADC, the used ADC is commonly SAR ADC or a flash ADC, which has a large area overhead and power consumption. The embedded column ADC was adopted to replace flash ADC and SAR ADC, which can alleviate the area overhead. The traditional column ADC needs fresh stored value "−1" or "+1" to alter the relative discharge between two bitlines during every cycle, which induces additional power consumption [see Fig. 1(c)] [15], [16].
3) The CIM macro mainly supports ternary activation and binary weights. The operations for ternary activation and ternary weights are still lack.
4) Realization of the Boolean logic operation requires an additional reference voltage generation circuit, which causes additional area and power consumption [see Fig. 1(e)].

To solve these issues, we propose a T8T-SRAM CIM macro, which possesses following features: 1) the proposed T8T-SRAM cell has separated read and write path, and can avoid read disturb issue with lower computing power consumption; 2) in Boolean logic operation mode, T8T-SRAM macro can achieve NAND, NOR, XNOR, and XOR operations with redundant rows, avoiding additional reference voltage generation circuit; 3) in MAC mode, the proposed CIM macro can achieve MAC operation of ternary activation and ternary weight, and the MAC result is readout in digital code by embedded column ADC; and 4) compared with other embedded column ADCs, the proposed column ADC does not need weight refresh during every cycle. The proposed embedded column ADC changes the discharge amount of computing bitlines by means of changing activations. And these operation modes are controlled by the mode switching circuit.

This work is organized as follows. Section II describes the CIM macro architecture and detailed circuit design. Section III presents simulation results and compares with state-of-the-art works. Section IV concludes this article.

II. Proposed SRAM CIM Macro

Fig. 2 presents the overall architecture of the proposed T8T-SRAM CIM macro, which mainly consists of a 32 × 32 8T-SRAM array, CIM control circuits, row/column decode circuits, pre-charge circuit, data input and driver circuits, and

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:37:55 UTC from IEEE Xplore. Restrictions apply.

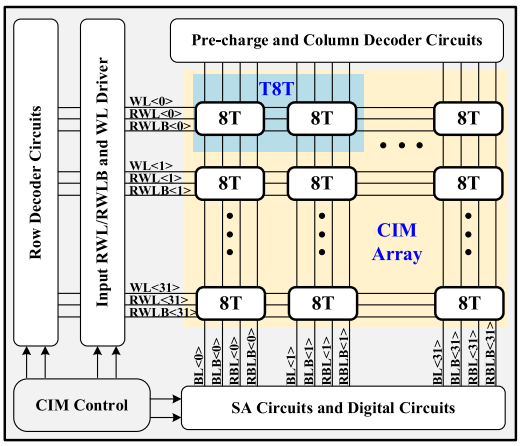
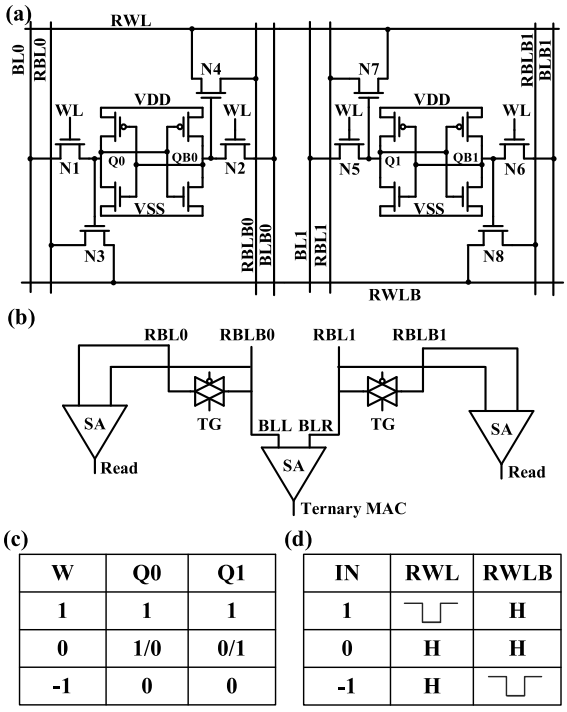


Fig. 2. Architecture of the proposed T8T-SRAM CIM macro.



(c)

W	Q0	Q1
1	1	1
0	1/0	0/1
-1	0	0

(d)

IN	RWL	RWLB
1	⎍	H
0	H	H
-1	H	⎍

Fig. 3. Circuit structure of (a) T8T-SRAM bitcell and (b) readout circuit. The truth table of (c) ternary weight and (d) ternary activation.

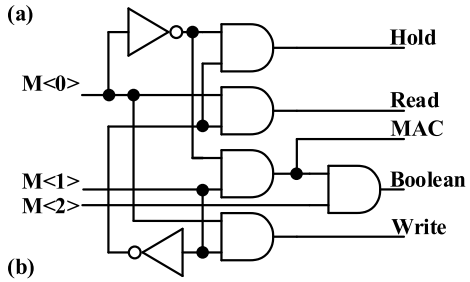
readout circuit. The T8T-SRAM CIM macro can be configured in SRAM mode, Boolean logic operation mode, and MAC mode. In SRAM mode, the T8T-SRAM macro works as a conventional SRAM array. In Boolean logic mode, the T8T-SRAM CIM macro can achieve NAND, NOR, XNOR, and XOR operations with SA and digital circuits. In MAC mode, the T8T-SRAM CIM macro can parallelly realize MAC operation of 16 ternary activations and 16 ternary weights.

A. Circuit Structure of T8T-SRAM Bitcell

The 8T-SRAM has two extra access transistors compared with the traditional 6T-SRAM cell, and can avoid the read disturb problem with separated read and write paths. However, the single 8T-SRAM can only store a 1-bit value. The 1-bit ternary value ("−1," "0," and "+1") needs to be stored by a new bitcell. Fig. 3(a) shows the circuit structure of the

TABLE I
CHARACTERISTIC COMPARISONS BETWEEN T8T-SRAM AND OTHER BITCELLS

	This work	[10]	[8]	[13]	[9]	[18]
Bitcell	T8T	10T	16T1C	19T	12T	8T
Technology	28 nm	28 nm	65 nm	32 nm	55 nm	7 nm
Read disturb	NO	NO	NO	NO	YES	NO
Ternary weight	YES	NO	YES	YES	YES	NO
Operation modes	SRAM/MAC /Boolean logic	SRAM /MAC	SRAM /MAC	SRAM /MAC	SRAM /MAC	SRAM /MAC



(b)

M<0:1> / M<2>	00	01	10	11
0	Hold	Read	MAC	Write
1	Hold	Read	Boolean	Write

Fig. 4. (a) Schematic of the mode-configured circuit and (b) truth table.

proposed T8T-SRAM bitcell, which consists of two 8T-SRAM bitcells. In T8T-SRAM bitcell, the added access transistors (N3, N4, N7, and N8) are connected with comping bilines (RBL0, RBLB0, RBL1, and RBLB1) and read bilines (RWL and RWLB). The 8T-SRAMs in a column share the BLi, BLBi, RBLi, and RBLBi bitlines ($0 \le i \le 31$), where BLi and BLBi are used to write operations and RBLi and RBLBi are used to read and MAC operations.

Fig. 3(b) shows the readout circuit of T8T-SRAM. Taking a T8T-SRAM column as an example, RBL0/RBL1 and RBLB0/RBLB1 are connected by transmission gates (TGs), and adjacent read bitlines RBLB0 and RBL1 are as inputs of SA. The TGs are disactivated when the CIM macro works in SRAM mode and Boolean logic mode. When the CIM macro works in MAC mode, the TGs are activated. The ternary value is the combination of $Q0$ and $Q1$ values. Also, the ternary activation is represented by a combination of voltage pulses of RWL and RWLB. The storage value truth table, i.e., the weight truth table and activation truth table, are shown in Fig. 3(c) and (d), respectively.

Among the previously reported bitcells for ternary weight implementation [8], [9], [13], the bitcells used in [8] and [13] suffer from low cell density and poor overall area efficiency, while [9] uses the dual-6T-SRAM bitcell with the read disturb problem. Jeong et al. [17] use the 3T1C-eDRAM bitcell to realize the MAC operation of ternary input and ternary weight. Compared with SRAM, 3T1C-eDRAM needs periodic refreshing. Table I shows the characteristic comparisons between T8T-SRAM and other bitcells.

As shown in Fig. 4, the configured circuit can set macro in SRAM hold, read, write modes, Boolean logic mode, and

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:37:55 UTC from IEEE Xplore. Restrictions apply.

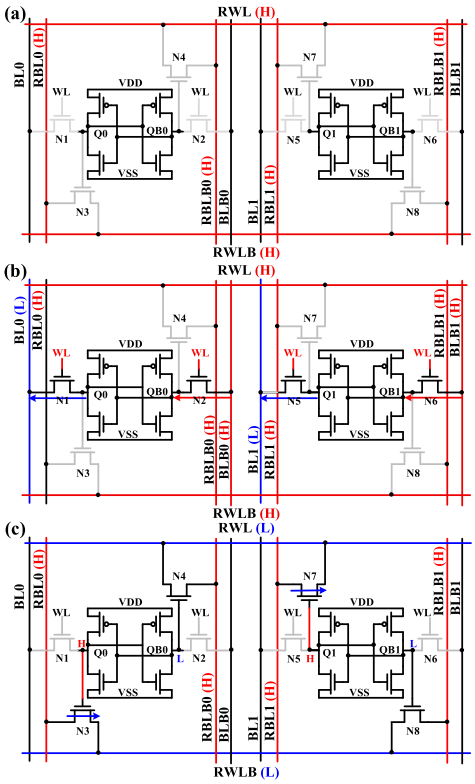


Fig. 5. Illustrations of (a) hold mode, (b) write mode, and (c) read mode of T8T-SRAM.

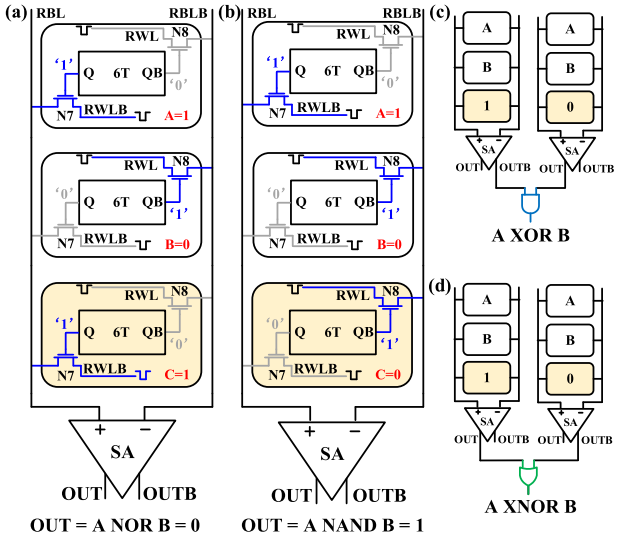


Fig. 6. Illustration of (a) NOR, (b) NAND, (c) XOR, and (d) XNOR logic operations in T8T-SRAM columns.

(a)

A	B	C	RBL	RBLB	OUT
0	0	1	1ΔV	2ΔV	1
0	1	1	2ΔV	1ΔV	0
1	0	1	2ΔV	1ΔV	0
1	1	1	3ΔV	0ΔV	0

(b)

A	B	C	RBL	RBLB	OUT
0	0	0	0ΔV	3ΔV	1
0	1	0	1ΔV	2ΔV	1
1	0	0	1ΔV	2ΔV	1
1	1	0	2ΔV	1ΔV	0

Fig. 7. Truth tables of (a) NOR logic and (b) NAND logic operations.

ternary MAC mode. Both of them are operated by controlling the states of RWL and RWLB in the corresponding bitcells. For ternary MAC operation, the RBL and RBLB are connected by TGs to form BLL and BLR bitlines, and the combination of RWL and RWLB is as input activation. For Boolean logic, the RBL and RBLB are disconnected by TGs. The RWL and RWLB of the two selected rows are both input activation with a low pulse, while the RWL and RWLB of the other rows remain high. The redundant row is preconfigured through a write operation to set the bitcell as "0" or "1."

B. Operations of T8T-SRAM Bitcell

In T8T-SRAM hold mode [see Fig. 5(a)], RBL0/RBLB0, RBL1/RBLB1, and RWL, RWLB are set as H; WLs are set as L. The storage node is isolated with an external circuit. In T8T-SRAM write mode [see Fig. 5(b)], RWL, RWLB, RBL0, RBLB0, RBL1, RBLB1 are set as H, and WL is set as H. Access transistors are opened, datum needed to write is performed on BL0/BLB0 and BL1/BLB1. Supposed "−1" ($Q0$ = "0" and $Q1$ = "0") is written in T8T-SRAM, the BL0, BL1 are set as "0," and BLB0, BLB1 are set as "1." In T8T-SRAM read mode [see Fig. 5(c)], WL, RWL, RWLB are set as L, RBL0/RBLB0, RBL1/RBLB1 are set as H. Due to the storage value, the RBL0 (RBL1) or RBLB0 (RBLB1) will discharge through access transistors. Supposed "1" ($Q0$ = "1" and $Q1$ = "1") is stored in T8T-SRAM, RBL0 and RBL1 will discharge though N3 and N7. With the help of SA, datum "1" is read out.

C. Boolean Logic Operations

In prior works, achieving Boolean logic operations in SRAM arrays requires additional reference voltage generation and assist circuits [19], [20], [21], which brings in additional power and area overhead. In the T8T-SRAM CIM macro, two rows of A and B are used as input data, and row C is used as a redundant cell for providing a reference voltage. With the help of redundant rows, the proposed T8T-SRAM CIM macro can save area and power consumption for Boolean logic operations.

Fig. 6 shows the illustration of Boolean logic operations. The logic inputs are performed in any two rows A and B in an 8T-SRAM column. The redundant row is C, which is used to provide a reference voltage. When performing NOR and NAND operations, RWL and RWLB are set as L, RBL and RBLB are set as H. As illustrated in Fig. 6(a), A is "1" (Q ="1," QB = "0"), B is "0" (Q = "0," QB = "1"), and C is "1" (Q = "1," QB = "0"). In this case, the RBL/RBLB are pre-charged to VDD at the beginning, then RBL discharges $2\Delta V$ and RBLB discharges $1\Delta V$, the NOR result of SA is 0. While in Fig. 6(b), A is "1," B is "0," and C is "0," the NAND result of SA is "1." The proposed circuit can also achieve XOR and XNOR logics by adding an AND gate and an OR gate, respectively [see Fig. 6(c) and (d)]. The truth table of NOR logic and NAND logic are listed in Fig. 7(a) and (b), respectively.

D. Ternary MAC Operations

For the ternary multiplication operation in a single T8T-SRAM, RBL and RBLB in a column are connected

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:37:55 UTC from IEEE Xplore. Restrictions apply.

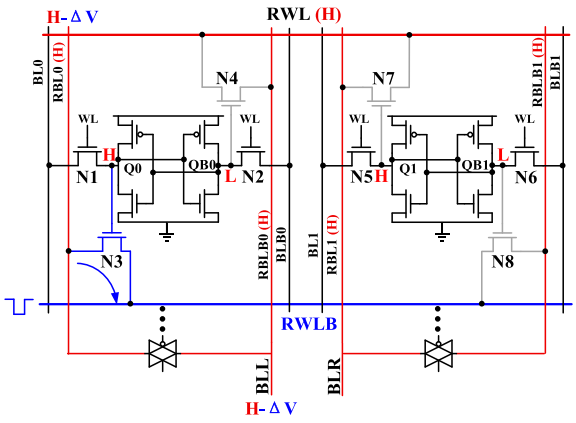


Fig. 8. Illustration of ternary multiplication operation [input (−1) × weight (1)].

TABLE II
TRUTH TABLE OF TERNARY MULTIPLICATION

W	Q0	Q1	IN	RWL	RWLB	OUT	ΔV_{BLL}	ΔV_{BLR}
1	1	1	1	⊓	H	1	0	-ΔV
0	1/0	0/1	1	⊓	H	0	0/-ΔV	0/-ΔV
-1	0	0	1	⊓	H	-1	-ΔV	0
X	/	/	0	H	H	0	0	0
1	1	1	-1	H	⊓	-1	-ΔV	0
0	1/0	0/1	-1	H	⊓	0	0/-ΔV	0/-ΔV
-1	0	0	-1	H	⊓	1	0	-ΔV

through TGs. The T8T-SRAM computing bitline on the left side is named BLL, and the right side is named BLR.

As shown in Fig. 3(c) and (d), the combinations of $Q0/Q1$ = "1"/"1," "0"/"0" represent weight = "1"/"−1," respectively. In other cases, the weight is "0." When input activation is "1," the RWL generates a low voltage pulse, and RWLB remains VDD (H). When input activation is "−1," RWLB generates a low voltage pulse, and RWL remains VDD. Both RWL and RWLB remain VDD, representing input activation is "0." Before the multiplication operation, BLL and BLR are precharged to VDD; then, BLL and BLR will discharge or not according to the multiplication operation between activation and weight. As illustrated in Fig. 8, when the activation is "−1," and the weight is "1," the BLL discharges $1\Delta V$, and the BLR remains at H. The final result is "−1." The ternary multiplication truth table is shown in Table II.

As illustrated in Fig. 9(a), two computing bitlines (BLL and BLR) of 16 rows are used for accumulating ternary multiplication results, and the other 16 rows are used for providing quantized reference voltage. Then, the final MAC results are quantized with the help of SA. The operation timing diagram is shown in Fig. 9(b). In the precharge phase, BLL and BLR are set to VDD. In the MAC phase, RWL⟨0:15⟩/RWLB⟨0:15⟩ are activated and used for multiplication operations, and the activation status of RWL is determined by the IN⟨0:15⟩. With the mapping relationship shown in Table I, the discharge amount of BLL and BLR is determined by the MAC results in a column. In this phase, RWL/RWLB⟨16:31⟩ remain unchanged and do not participate in the operation of MAC. Assuming the number of multiplication results "1" is five rows, then the BLR will discharge $5\Delta V$, and if the number of multiplication results "−1" is seven rows, then the BLL will discharge $7\Delta V$. After the MAC phase, the calculated results are quantified with the help of 16 reference rows and SA. The MAC and quantization phases are performed in parallel, which are separated in the timing diagram.

E. Proposed Embedded 4-Bit Column ADC

In this work, we use an embedded SRAM array to convert analog voltage to digital output. The quantization result of MAC is represented by multicycle SA output thermometer coding. In previous studies on embedded column ADCs (see Fig. 10) [15], [16], the stored values of "−1" or "1" are changed to alter the relative voltages on two bitlines. With the comparison of SA, the MAC results are approaching. Switching the reference value between "−1" and "+1" requires modifying the weight values, which leads to significant power consumption and latency.

Fig. 11(a) shows the operating principle of the proposed embedded 4-bit column ADC. The redundant rows scan the value from −14 to +14 in 15 cycles (every scan step is "2"). As shown in Table II, the discharge of ΔV_{BLL} and ΔV_{BLR} is determined by the MAC results of T8T-SRAM bitcells. The amounts of ΔV_{BLL} and ΔV_{BLR} can be adjusted by changing RWL and RWLB, without refreshing the stored values. Through more bitcell reconfiguration, the column ADC can also support high bits.

As shown in Fig. 11(b), the MAC result and reference voltage are compared in every cycle. The reference voltage is presented by the discharge amount of BLL and BLR, and this discharge amount is superimposed with the MAC discharge amount. Suppose the ternary MAC result is "−1" (BLL has a discharge amount of ΔV, while BLR remains VDD), and the result remains unchanged in every quantitative cycle. In cycle 1, the number of "−1" is 15, and "+1" is 1 in redundant rows. In this case, the total discharge amount of BLL is $16\Delta V$, and BLR is $1\Delta V$. The BLL voltage is less than BLR; therefore, the output of SA is 0 (TH⟨1⟩ = 0). In cycle 2, based on the analysis of cycle 1, the output of SA is 0 (TH⟨2⟩ = 0). In this way, we can obtain the comparison results of SA in all cycles, and combine the results to obtain the thermometer code (000_0000_0111_1111). Fig. 11(c) shows the corresponding thermometer code of the MAC result. The final result is realized by converting the thermometer code to a 4-bit binary code (0111). Compared with prior CIM architectures [22], [23], the proposed 4-bit column ADC reduces additional ADC or other assist circuits. In the prior column, ADC works [15], [16], the input value is binary (0, 1), and switching the reference value between "−1" and "+1" requires modifying the weight values, which leads to significant power consumption and latency. The proposed 4-bit column ADC does not need to rewrite the weight data during different cycle operations. It saves additional area overhead and power consumption. Meanwhile, the SRAM

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:37:55 UTC from IEEE Xplore. Restrictions apply.

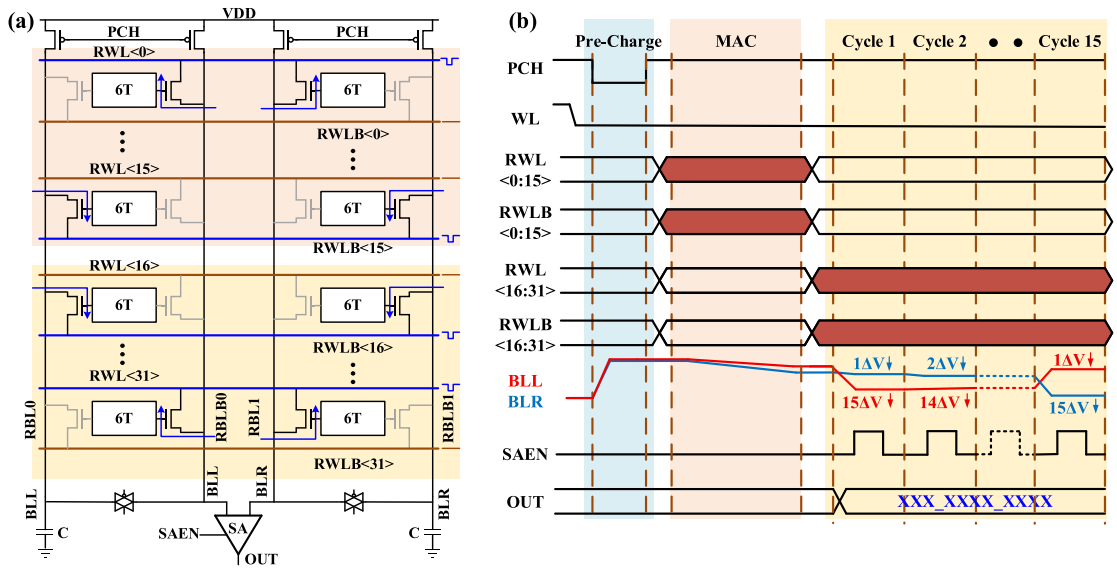


Fig. 9. Illustrations of (a) multirow MAC operation and (b) operation waveform in different phases.

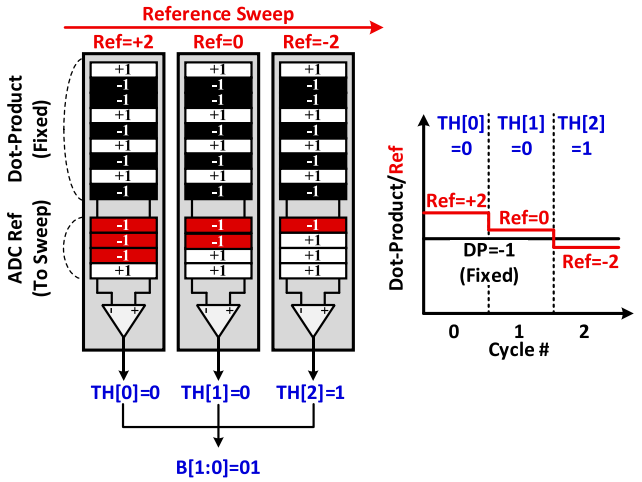


Fig. 10. Operating principle of the previously proposed column ADC [15].

used in the proposed ADC architecture can be utilized for both SRAM mode operation and Boolean logic operations, reducing area overhead. Considering the scalability, the proposed ADC architecture can also be comparable with column-shared SAR ADC, due to the large area of the CDAC circuit.

III. Simulation Results and Analysis

The proposed CIM macro was simulated using 28-nm CMOS technology. To evaluate the performance of T8T-SRAM read ability, simulations of static noise margin (SNM) and read speed were performed. As shown in Fig. 12(a), we compare the read SNM (RSNM) of traditional T8T-SRAM and 6T-SRAM in the range of 0.5–0.9 V. In the condition of a wide range of supply voltages, the RSNM of T8T-SRAM is larger than that of 6T-SRAM. In typical conditions (TT corner, 27 °C, and 0.9 V), the RSNM of T8T-SRAM is 317.5 mV, which is ~3.3 times higher than that of 6T-SRAM (95.1 mV). As shown in Fig. 12(b), the read speed of T8T-SRAM and 6T-SRAM under different supply voltages and different temperatures is performed. In different cases, the read speed of T8T-SRAM is faster than that of 6T-SRAM, and is more robust to temperature variations.

Fig. 13(a) shows the circuit structure of the voltage-latch type SA. 2k samples of Monte Carlo simulations show that the standard deviation (σ) of offset voltage is 2.95 mV (see Fig. 13(b), TT corner, 27 °C, and 0.9 V). When the input voltage is larger than 8.78 mV ($\mu \pm 3\sigma$), the SA will give correct results.

To verify the feasibility of Boolean logic operation, Fig. 14(a) shows the transient waveform of NOR operation, taken "1 NOR 0" as an example ($C = 1$). The discharge amount of RBL is more than that of RBLB, so the SA output is "0." 2k samples of Monte Carlo simulation were performed to evaluate the minimum difference voltage (ΔV) between the RBL and RBLB at 0.5-V array supply voltage and 0.9-V peripheral supply voltage [see Fig. 14(b)]. The voltage difference distribution shows that the ΔV between 23.82 and 100.04 mV is 99.7%. The ΔV is much higher than 6σ of the SA offset voltage, which ensures the correct Boolean logic operation.

For ternary MAC operations, the accuracy is expressed by integral nonlinearity (INL). The smaller the INL value, the better the linearity and accuracy of the computation result. Fig. 15(a) and (b) shows the postlayout voltage change of computing bitline and simulated INL of ternary MAC operation with different MAC values. The postlayout simulations have four cases: $R + C+$ CC, $C+$ CC, $R + C$, and R only (R represents THE resistance, C represents the capacitance, and CC represents the coupling capacitance). It shows that the voltage change is largest in the case of resistance only (R). This is because the capacitor has charge storage ability. When the MAC result is 16 in this case ($R + C+$ CC), the INL value reaches its maximum, which is less than 1.21 LSB. Compared with [24] and [25], the proposed T8T-SRAM array has a better performance on linearity due to the separated read path.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:37:55 UTC from IEEE Xplore. Restrictions apply.

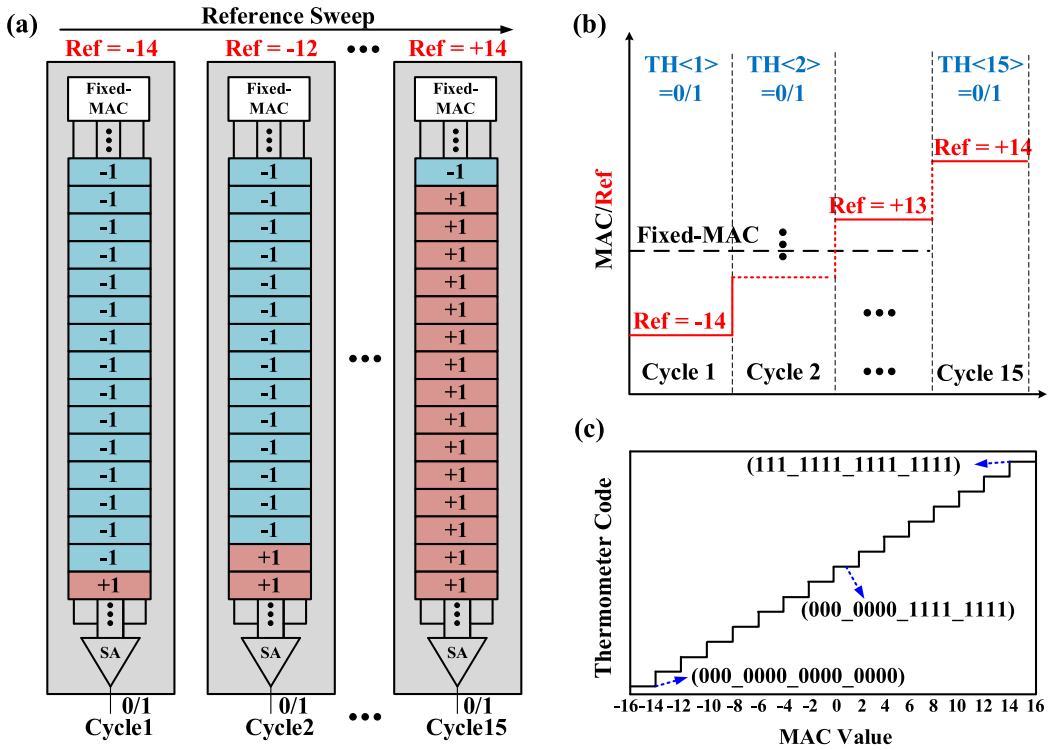


Fig. 11. (a) Quantization scheme of the proposed embedded 4-bit column ADC. (b) Timing diagram of quantization operations. (c) Corresponding thermometer code of the MAC result.

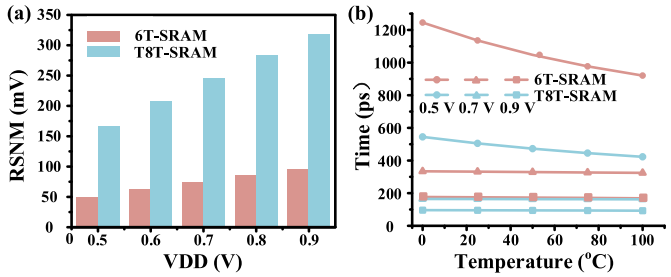


Fig. 12. Comparison of (a) RSNM and (b) read speed of T8T-SRAM and 6T-SRAM.

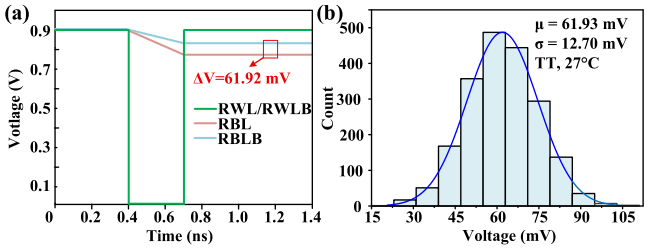


Fig. 14. (a) Transient waveform of "1 NOR 0." (b) 2k samples of the corresponding Monte Carlo simulation.

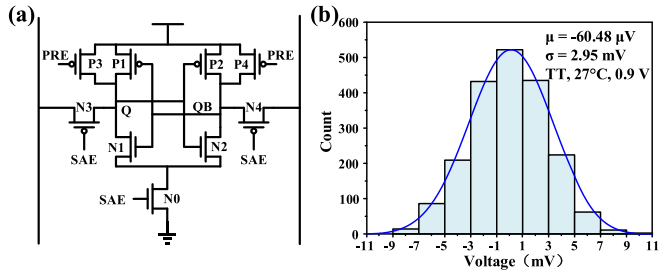


Fig. 13. (a) Circuit structure of voltage-latch type SA and (b) 2k samples of Monte Carlo simulation.

Fig. 15(c) and (d) shows the postlayout simulated power and energy efficiency of ternary MAC operation at different MAC values (−16 to +16). The power and energy efficiency of the negative part and the positive part of the MAC value are nearly symmetrical. In Fig. 15(c), when the MAC value is "0," the computing is performed with all "0" activation. Thus, there is no discharge path, resulting in a minimum power of 48.02 μW with only the R case. When the MAC value is ± 16, each computing row has a discharge path. In the $R + C$ case, the power is 84.98 μW. The maximum power occurs under two scenarios: weight W = "0" (Q"0"/Q"1" = "0"/ "1")/(Q"0"/"1" = "1"/"0") and activation = "1" (a low pulse is applied to RWL)/activation = "−1" (a low pulse is applied to RWLB). In these scenarios, both BLL and BLR have discharge paths. In a typical case ($R + C$+ CC), the maximum power is 117.8 μW. Also, the evaluated energy efficiency of MAC operation is 333.99–816.1 TOPS/W.

Fig. 16 illustrates how the activation ($3 \times 3 \times 16$) and weight kernel ($3 \times 3 \times 16 \times 16$) of the convolution are mapped to the proposed CIM macro. The weight kernel is unrolled and grouped into nine 16×16 weight matrices. The inputs are grouped into 9×16 matrices. For the macro limitation, the MAC operation is suit for convolutional layer and fully connected layer in CNNs and ResNets. While other layers need to be postprocessed in the digital domain circuit.

Fig. 17(a) shows the layout of the T8T-SRAM macro, and the size is 107.04×98.165 μm. Fig. 17(b) shows the area breakdown of the T8T-SRAM macro, where the readout circuit

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:37:55 UTC from IEEE Xplore. Restrictions apply.

TABLE III
FEATURE SUMMARY AND COMPARISON WITH RELATED WORKS

	This work	TCAS II'23 [8]	VLSI'20 [13]	TCAS I'23 [10]	SSCL'25 [26]	TCAS II'24 [27]
Technology	28 nm	65 nm	32 nm	28 nm	28 nm	28 nm
Array size	1 Kb	9.22 KB	4 Kb (block)	16 Kb	40 Kb	8 Kb
Bitcell type	T8T	16T1C	19T	10T	SLC-RRAM	10T
Input	ternary	ternary	ternary	ternary	binary	5 bit
Weight	ternary	ternary	ternary	binary	ternary	ternary
Energy efficiency (TOPS/W)	333.99-816.1	823	132.7	942.9-2941	1250.5	1414.9
Area efficiency (TOPS/mm^2)	61.9	NA	30.6	59.584	0.367	102.069

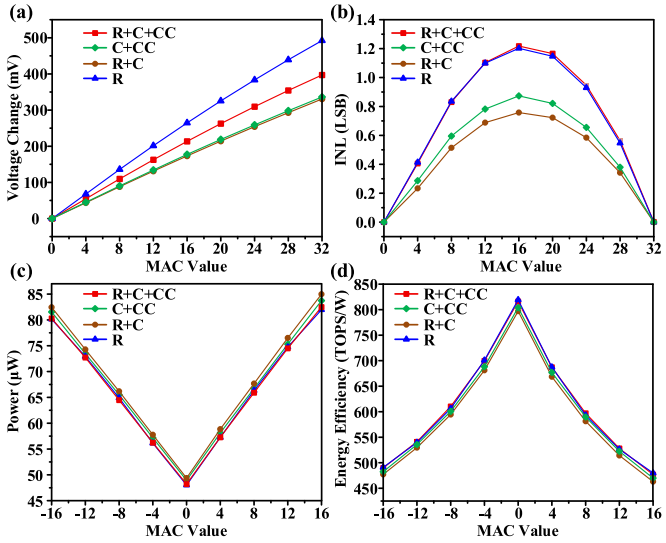


Fig. 15. (a) Voltage change of computing bitline and (b) INL with different MAC results. (c) Power and (d) energy efficiency plots with different MAC values. *R* represents the resistance, *C* represents the capacitance, and CC represents the coupling capacitance.

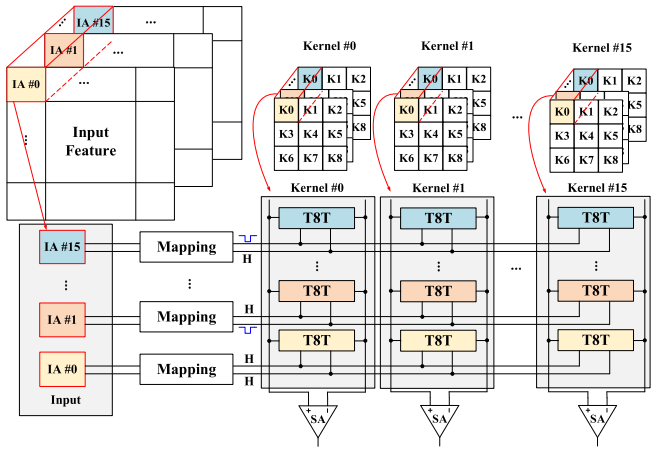


Fig. 16. Mapping input feature maps and kernels of the convolutional layer to T8T-SRAM CIM macro.

(SA and digital circuit) occupies 23.7% of the area; the 32×32 SRAM array occupies 24.5% of the area; the mode switching and control circuit occupies 2.4% of the area.

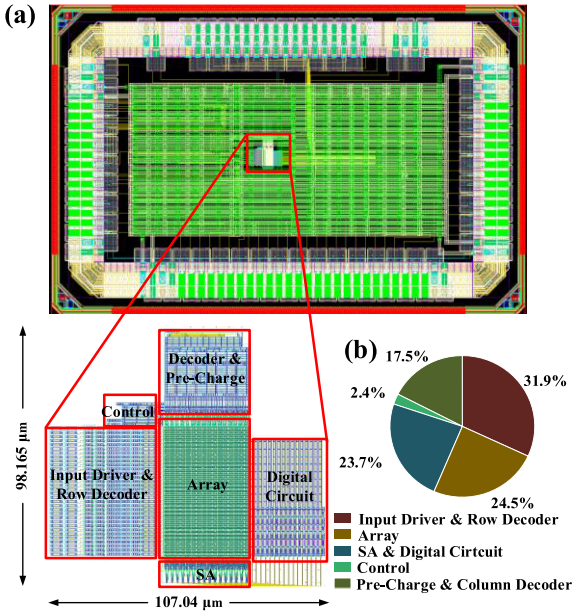


Fig. 17. (a) Layout of T8T-SRAM CIM macro. (b) Area breakdown of T8T-CIM macro.

Table III lists the comparison of its characteristics with previous SRAM-based CIM works. The proposed T8T-SRAM CIM macro can perform MAC operation between ternary activation and ternary weight with 333.99–816.1-TOPS/W energy efficiency and 61.9 TOPS/mm^2 area efficiency with parallel operations under 0.5-V array supply voltage and 0.9-V peripheral supply voltages. The area efficiency of the proposed T8T SRAM macro is better than that of RRAM and a hybrid combination of 19T SRAM. Since the small memory array capacity (1 kb), the array accounts for a lower area proportion, thereby reducing the overall area efficiency. Compared with analogous ternary input and ternary weight CIM macros [8], [13], the T8T-SRAM CIM macro has less transistors in the bitcell, and shows better performance in energy efficiency. With the same technology node and equivalent array capacity, the proposed T8T SRAM macro still performs better area efficiency compared with other architectures [10], [27]. Using a ResNet-18 network for the inference of MNIST and CIFAR-10 datasets, the accuracies were 99.06% and 85.76% with a ternary input and ternary weights.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:37:55 UTC from IEEE Xplore. Restrictions apply.

IV. Conclusion

This work proposed a T8T-SRAM macro, which can be configured in SRAM, Boolean logic, and ternary MAC operations. The proposed T8T-SRAM bitcell shows high RSNM and temperature stability. In Boolean logic operation mode, T8T-SRAM CIM macro can achieve NAND, NOR, XNOR, and XOR operations with redundant rows, avoiding additional reference voltage generation and refresh circuit. 2k samples of Monte Carlo simulation show the feasibility of Boolean logic operations. In MAC operation, when 16 rows are used for embedded column ADC, T8T-SRAM CIM macro can parallelly realize the MAC operation of 16 ternary activations and 16 ternary weights. In 28-nm CMOS technology, under 0.5-V array supply voltage and 0.9-V peripheral supply voltage, the postlayout simulations manifest the INL of MAC is less than 1.21 LSB. When the T8T-SRAM CIM performs MAC operation of 16 ternary activations and 16 ternary weights, it shows 333.99–816.1-TOPS/W energy efficiency and 61.9-TOPS/mm^2 area efficiency. Using an ResNet-18 network for the inference of MNIST and CIFAR-10 datasets, the accuracies were 99.06% and 85.76% with a ternary input and ternary weights. The T8T-SRAM circuit is competitive among the state-of-the-art CIM macro circuits; it provides an attractive candidate for in-memory acceleration for edge computing.

References

[1] D. Zou, G. Zhang, X. Zhang, M. Wang, and Z. Wang, "An efficient and precision-reconfigurable digital CIM macro for DNN accelerators," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 33, no. 2, pp. 563–567, Feb. 2025.

[2] H. Hu et al., "A 40-nm SONOS digital CIM using simplified LUT multiplier and continuous sample-hold sense amplifier for AI edge inference," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 31, no. 12, pp. 2044–2052, Dec. 2023.

[3] M. A. Zidan, J. P. Strachan, and W. D. Lu, "The future of electronics based on memristive systems," *Nature Electron.*, vol. 1, pp. 22–29, Jan. 2018.

[4] X. Xiang, Z. Yue, X. Zhang, S. Wei, Y. Hu, and S. Yin, "Dyn-bitpool: A 28 nm 27 TOPS/W two-sided sparse CIM accelerator featuring a balanced workload scheme and high CIM macro utilization," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 72, no. 5, pp. 2216–2228, May 2025.

[5] H. Jia, H. Valavi, Y. Tang, J. Zhang, and N. Verma, "A programmable heterogeneous microprocessor based on bit-scalable in-memory computing," *IEEE J. Solid-State Circuits*, vol. 55, no. 9, pp. 2609–2621, Sep. 2020.

[6] S. Okumura, M. Yabuuchi, K. Hijioka, and K. Nose, "A ternary based bit scalable, 8.80 TOPS/W CNN accelerator with many-core processing-in-memory architecture with 896K synapses/mm2," in *Proc. Symp. VLSI Technol.*, Jun. 2019, pp. C248–C249.

[7] Y. Wang, F. Tu, L. Liu, S. Wei, Y. Xie, and S. Yin, "SPCIM: Sparsity-balanced practical CIM accelerator with optimized spatial–temporal multi-macro utilization," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 70, no. 1, pp. 214–227, Jan. 2023.

[8] H. Jeong, S. Kim, K. Park, J. Jung, and K. J. Lee, "A ternary neural network computing-in-memory processor with 16T1C bitcell architecture," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 70, no. 5, pp. 1739–1743, May 2023.

[9] X. Wang, G. Li, J. Sun, H. Fan, Y. Chen, and H. Jiao, "Ternary in-memory MAC accelerator with dual-6T SRAM cell for deep neural networks," in *Proc. IEEE Asia–Pacific Conf. Circuits Syst. (APCCAS)*, Nov. 2022, pp. 246–250.

[10] S. Cheon, K. Lee, and J. Park, "A 2941-TOPS/W charge-domain 10T SRAM compute-in-memory for ternary neural network," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 70, no. 5, pp. 2085–2097, Feb. 2023.

[11] K. Ando et al., "BRein memory: A single-chip binary/ternary reconfigurable in-memory deep neural network accelerator achieving 1.4 TOPS at 0.6 W," *IEEE J. Solid-State Circuits*, vol. 53, no. 4, pp. 983–994, Apr. 2018.

[12] S. Yin, Z. Jiang, J.-S. Seo, and M. Seok, "XNOR-SRAM: In-memory computing SRAM macro for binary/ternary deep neural networks," *IEEE J. Solid-State Circuits*, vol. 55, no. 6, pp. 1733–1743, Jun. 2020.

[13] S. Jain, S. K. Gupta, and A. Raghunathan, "TiM-DNN: Ternary in-memory accelerator for deep neural networks," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 28, no. 7, pp. 1567–1577, Jul. 2020.

[14] C. Yu, H. Jiang, J. Mu, K. T. C. Chai, T. T.-H. Kim, and B. Kim, "A dual 7T SRAM-based zero-skipping compute-in-memory macro with 1-6b binary searching ADCs for processing quantized neural networks," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 8, pp. 3672–3682, Aug. 2024.

[15] C. Yu, T. Yoo, K. T. C. Chai, T. T. Kim, and B. Kim, "A 65-nm 8T SRAM compute-in-memory macro with column ADCs for processing neural networks," *IEEE J. Solid-State Circuits*, vol. 57, no. 11, pp. 3466–3476, Nov. 2022.

[16] C. Dai et al., "A 9T-SRAM in-memory computing macro for Boolean logic and multiply-and-accumulate operations," *Microelectron. J.*, vol. 144, Feb. 2024, Art. no. 106087.

[17] H. Jeong, S. Kim, J. Shin, and K. Jason Lee, "HYTEC: Compact and energy-efficient analog-digital hybrid CIM with transpose ternary eDRAM," *IEEE J. Solid-State Circuits*, early access, 2025, doi: 10.1109/jssc.2025.3623671.

[18] M. E. Sinangil et al., "A 7-nm compute-in-memory SRAM macro supporting multi-bit input, weight and output and achieving 351 TOPS/W and 372.4 GOPS," *IEEE J. Solid-State Circuits*, vol. 56, no. 1, pp. 188–198, Jan. 2021.

[19] C. Moorthii, A. Singla, and M. Suri, "DNA-CIM: DNA sequence analysis using RRAM-based compute in-memory accelerator," in *Proc. 38th Int. Conf. VLSI Design 23rd Int. Conf. Embedded Syst. (VLSID)*, Jan. 2025, pp. 314–319.

[20] D. Challagundla, I. Bezzam, B. Saha, and R. Islam, "Resonant compute-in-memory (rCIM) 10T SRAM macro for Boolean logic," in *Proc. IEEE 41st Int. Conf. Comput. Design (ICCD)*, Nov. 2023, pp. 110–117.

[21] A. K. Rajput and M. Pattanaik, "Implementation of Boolean and arithmetic functions with 8T SRAM cell for in-memory computation," in *Proc. Int. Conf. Emerg. Technol. (INCET)*, Jun. 2020, pp. 1–5.

[22] S. Dou, Z. Gu, H. You, Y. Zhan, S. Qiao, and Y. Zhou, "A charge domain SRAM computing-in-memory macro with quantized interval-optimized ADC and input bit-level sparsity-optimized P2O-DAC for 8-b MAC operation," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 33, no. 5, pp. 1467–1471, May 2025.

[23] H. Zhang et al., "SSM-CIM: An efficient CIM macro featuring single-step multi-bit MAC computation for CNN edge inference," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 70, no. 11, pp. 4357–4368, Nov. 2023.

[24] J. Zhang et al., "In-memory multibit multiplication based on bitline shifting," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 2, pp. 354–358, Feb. 2022.

[25] W. Lu et al., "High-reliability and high-throughput CIM 10T-SRAM for multiplication and accumulation operations with 274.3 GOPS and 200–237.5 TOPS/W," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 33, no. 4, pp. 1073–1081, Apr. 2025.

[26] S. Wei et al., "A 28-nm static-power-free fully parallel RRAM-based TD CIM macro with 1982 TOPS/W/bit for edge applications," *IEEE Solid-State Circuits Lett.*, vol. 8, pp. 21–24, 2025.

[27] K. Xiao et al., "A 28nm 8Kb reconfigurable SRAM computing-in-memory macro with input-sparsity optimized DTC for multi-mode MAC operations," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 7, pp. 3263–3267, Jul. 2024.



Chenghu Dai (Member, IEEE) received the Ph.D. degree from the School of Microelectronics, Shandong University, Jinan, China, in 2019.

He is currently a Lecturer of Microelectronics and Solid-State Electronics with Anhui University, Hefei, China. His current research interests include high-performance and reliable memory design, in memory-computing, and analog IC design.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:37:55 UTC from IEEE Xplore. Restrictions apply.



Jianhao Zhang received the B.S. degree in microelectronics science and engineering from Anhui University, Hefei, China, in 2023, where he is currently working toward the M.S. degree at the School of Integrated Circuits.

His research interests include computing-in-memory and digital circuit design.



Zihua Ren received the B.S. degree in communications engineering and the M.S. degree in integrated circuit engineering from Anhui University, Hefei, China, in 2022 and 2025, respectively.

His research interests include nonvolatile memory design and computing-in-memory.



Lian Liu received the B.S. degree in integrated circuit design and integrated systems from Chongqing University of Posts and Telecommunications, Chongqing, China, in 2024. He is currently working toward the M.S. degree at the School of Integrated Circuits, Anhui University, Hefei, China.

His research interests include computing-in-memory and low-power circuit design.



Lei Meng received the B.S. degree in microelectronics science and engineering from Hefei Normal University, Hefei, China, in 2024. He is currently working toward the M.S. degree at the School of Integrated Circuits, Anhui University, Hefei.

His current research interests include SRAM circuit design and memory performance improvement.



Chaoyi Wang received the B.S. degree in microelectronics science and engineering from Ningbo University, Zhejiang, China, in 2024. He is currently working toward the M.S. degree at the School of Integrated Circuits, Anhui University, Hefei, China.

His current research interests include high-performance SRAM design.



Chunyu Peng (Senior Member, IEEE) received the M.S. degree in circuits and systems and the Ph.D. degree in microelectronics and solid-state electronics from Anhui University, Hefei, China, in 2013 and 2022, respectively.

He is currently an Associate Professor of Microelectronics and Solid-State Electronics with Anhui University. His research interests include signal processing, analog IC design, high performance, and reliable memory technology.



Wenjuan Lu (Member, IEEE) received the M.S. degree in circuits and systems and the Ph.D. degree in microelectronics and solid-state electronics from Anhui University, Hefei, China, in 2014 and 2017, respectively.

She is currently an Associate Professor of Microelectronics and Solid-State Electronics with Anhui University. Her current research interests include next-generation nonvolatile memory, and the circuit design of low-power, high-performance, and high-reliability nonvolatile memory devices.



Zhiting Lin (Senior Member, IEEE) received the B.S. and Ph.D. degrees in electronics and information engineering from the University of Science and Technology of China (USTC), Hefei, China, in 2004 and 2010, respectively.

From 2015 to 2016, he was a Visiting Scholar with the Engineering and Computer Science Department, Baylor University, Waco, TX, USA. In 2011, he joined the School of Integrated Circuits, Anhui University, Hefei, where he is currently a Professor. He has published about 50 articles and holds over 20 Chinese patents. His research interests include pipeline analog-to-digital converters and high-performance static random access memory.



Xiulong Wu (Senior Member, IEEE) received the B.S. degree in computer science from the University of Science and Technology of China (USTC), Hefei, China, in 2001, and the M.S. and Ph.D. degrees in electronic engineering from Anhui University, Hefei, in 2005 and 2008, respectively.

From 2013 to 2014, he was a Visiting Scholar with the Engineering Department, The University of Texas at Dallas, Richardson, TX, USA. He is currently a Professor with Anhui University. He has published about 60 articles and holds over ten Chinese patents. His research interests include high-performance static random access memory and mixed-signal ICs.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 29,2026 at 13:37:55 UTC from IEEE Xplore. Restrictions apply.